

Ahmed Abdelkawy

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Summary

PhD Candidate in Electrical and Computer Engineering with expertise in object detection, multimodal action recognition, VLM/VLT/VLA models, and transformer-based architectures for 2D-to-3D multi-camera Bird's Eye View (BEV) object detection. Passionate about designing and optimizing deep learning models for physical AI, particularly autonomous driving.

Work Experience

Research Assistant, CVIP Lab, University of Louisville – Louisville, KY May 2022 – Present

- Proposed a novel multimodal deep learning architecture for human action recognition, achieving up to 72.8× reduction in FLOPs while maintaining competitive performance compared to the state-of-the-art methods.
- Introduced a novel framework for quantifying student behavioral engagement using a histogram of actions generated from the student's actions and gaze vectors.
- Designed a novel peer-aware student behavioral engagement estimation framework by fine-tuning VIFI-CLIP for student in-class action recognition and integrating Large Language Models (LLMs) for context-based reasoning over action sequences.
- Designed and evaluated efficient Vision Transformer architectures, exploring alternatives to quadratic self-attention through CLS-guided communication and spatial shifting on ImageNet-1K.
- Explored transformer-based BEV architectures (e.g., BEVFormer) for camera-based 3D perception and Vision-Language-Action (VLA) models (e.g., OpenVLA).
- Fine-tuned Large Vision Models, e.g., Qwen2.5 VL, for domain-specific video understanding.

Computer Vision Engineer, SawaSports Analytics startup – Cairo, Egypt Sept 2018 - Mar 2020

- Developed a multi-camera tracking system integrating a YOLO detector and DeepSORT for detecting and tracking players across overlapping camera views, projecting player locations into a unified bird's-eye-view (BEV) representation of the playground.
- Implemented a convolutional neural network-based player identification model via jersey number recognition.
- Built a Finite State Machine (FSM)-based approach for recognizing key game events, e.g., passes, shots on goal.
- Developed a skill assessment framework integrating CenterNet for joint player detection and pose estimation with biomechanical motion analysis to assess skill execution and generate actionable feedback.

Research Assistant, Helwan University – Cairo, Egypt Oct 2016 - Aug 2018

- Developed unsupervised machine learning techniques for anomaly detection in crowded pedestrian scenes, achieving real-time performance (25 fps) and high accuracy (98.48% AUC on UMN dataset).

Technical Skills

Programming Languages: Python, Java, C/C++, Matlab, CUDA C. **Tools:** Git, Claude Code.

Libraries/Frameworks: PyTorch, Tensorflow/Keras, OpenCV, NumPy, Pandas.

Education

Ph.D. in Electrical and Computer Engineering, University of Louisville, KY, USA. May 2022 – Present

- Thesis: Multimodal Learning for Human Action Recognition with Application to Student Engagement Analysis.
- GPA: 3.9/4.0 | Expected Graduation: December 2026

M.Sc. in Computer Science (Computer Vision), Helwan University, Cairo, Egypt. Oct 2013 - Oct 2018

- Thesis:** Real-time Abnormal Behavior Detection in Ubiquitous Environment

BSc in Computer Science, Helwan University, Cairo, Egypt. GPA: 3.8/4.0 Sept 2008 - May 2012

Selected Publications

A. Abdelkawy, A. Elsayed, A. Ali, A. Farag, T. Tretter, M. McIntyre. "Context Matters: Peer-Aware Student behavioral Engagement Measurement via VLM Action Parsing and LLM Sequence Classification." CVPR. 2026

A. Abdelkawy, A. Ali, A. Farag. "EPAM-Net: An Efficient Pose-driven Attention-guided Multimodal Network for Video Action Recognition." Neurocomputing. 2025

A. Abdelkawy, A. Farag, I. Alkabbany, A. Ali, C. Foreman, T. Tretter, N. Hindy. "Measuring Student Behavioral Engagement using Histogram of Actions." Pattern Recognition Letters. 2024

Academic Service

Reviewer: IJCV | Pattern Recognition | Neurocomputing | Pattern Recognition Letters.